Sivakorn Khundilokrattaya

3rd-Year Computer Engineering Student · Seeking Data Science, ML & Quantitative Fintech Internship

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EDUCATION

2024 – Present
Expected 2028
Bangkok, Thailand

Srinakharinwirot University (SWU) – B.Eng. in Computer Engineering

Statistical ML, Credit Risk & Data Engineering · Coursework: ML, Statistics, Databases, OS

- **NHK Spring Scholar (2026–2027):** Conferred a 30,000 THB/year engineering award (Sep 2026, GPAX ≥ 2.75) with a 2-month summer internship.
- **Verified Benchmarks (2026):** Scored 97/150 on SKILLKAMP by KBTG Data Analyst STAMP (vs. 90 benchmark) and CEFR B1 (51/120) on Oxford Placement.

SKILLS

ML & Risk	LightGBM, CatBoost, XGBoost, Scikit-Learn, PyG (R-GCN), Optuna
XAI & Data	OptBinning, TreeSHAP, DiCE, Python (Pandas, NumPy, SciPy), SQL
Engineering	FastAPI, ONNX, Docker, WebSockets, PostgreSQL, React, Next.js, Git
Languages	Thai (Native), English (CEFR B1 – Oxford), Japanese (Elementary)

PROJECTS

2026
KBTG Top 10 DS

K-NEXUS – FlowSense & TrustGraph (Synthetic Prototype)

App / Showcase

PyTorch Geometric (R-GCN), LightGBM, ONNX Runtime, FastAPI, React 18

- Verified that a non-graph baseline (0.996 PR-AUC) beat 16D R-GCN + LightGBM embeddings (0.994 PR-AUC) across 1,354 chronological transfers.
- Kept 30-day direct-ML cashflow forecasting disabled after paired evaluation on 32 synthetic accounts (2,701.6 vs. 313.3 THB baseline MAE).

2026
Research Prototype

Credit Risk Scoring & Explainable AI (XAI) Decision Prototype

GitHub

Python, LightGBM, Optuna, OptBinning (WoE), TreeSHAP, DiCE, FastAPI, Next.js

- Raised cross-validation ROC-AUC from 0.742 to 0.785 with 43.2% KS separation by pairing monotonic WoE screening with 5-fold Optuna LightGBM.
- Mapped default probabilities into a 300–850 underwriting scorecard (Base 600, PDO 20) served in FastAPI/Next.js with TreeSHAP and DiCE.

2026
Shopee Challenge

Shopee Smart COD Reliability & Adaptive Checkout Prototype

GitHub

Python, Scikit-Learn (Random Forest), Pandas, FastAPI, JavaScript

- Reached 0.840 ROC-AUC and 0.306 PR-AUC on 3,500 synthetic holdout orders across 50,000 simulated checkouts using a 3-fold Random Forest.
- Projected 3.58M THB/month in reverse-logistics savings (at 58% RTO reduction) by mapping risk scores to a 5–99 scale and 4 checkout tiers.

2025
Aihack Finalist

Long Overdue Debtor (LOD) Default Prediction Pipeline

GitHub

CatBoost, LightGBM, XGBoost, Optuna, Scikit-Learn

- Advanced out-of-fold ROC-AUC from 0.633 to 0.653 across 1:12 imbalanced consumer loans (~8.3% default rate) via 5-fold rank-blended trees.
- Engineered leak-free credit utilization, tenure, and missingness features while publishing a synthetic pipeline for reproducible review.

MORE PROJECTS

2026
IoT Prototype

Industrial IoT Telemetry & OEE Data Platform (Simulated Telemetry) GitHub

Python, FastAPI, PostgreSQL 15, WebSockets, LightGBM, TreeSHAP, React 18

- Streamed live machine-health readings and hourly OEE summaries (Availability × Performance × Quality) via FastAPI, PostgreSQL, and WebSockets.
- Surfaced interpretable equipment anomaly flags on a React 18 dashboard by combining Isolation Forest, LightGBM, and TreeSHAP attributions.

2026
Official SWU Portal

SWU Open House 2026 Web Platform Enhancement Case Study

WordPress, Elementor, Custom CSS, Responsive UI, THE CONCERT Integration

- Streamlined university event onboarding into a guided 3-step registration flow integrated with THE CONCERT ticketing platform across devices.
- Built an interactive multi-faculty poster showcase and live countdown interface with responsive custom CSS and unified navigation.

Jan 2026
True Launchpad

GuardianAI Clinical Frailty Triage Dashboard (Prototype) GitHub

React, Vite, Node.js, Tailwind CSS, Thai / English Localization

- Unified patient screening registries, vital trends, and alert queues into a bilingual (Thai/English) care-coordination triage prototype.
- Structured clinical follow-up workflows on example patient records by linking risk-factor views with provider assignment and care notes.

Jan 2026
GISTDA Hackathon

30m Grid Flash-Flood Hazard Mapping Pipeline

Python, GeoPandas, Rasterio, Shapely, Sentinel-1 SAR, Copernicus DEM

- Segmented flood extent from dual-polarization Sentinel-1 SAR backscatter and Copernicus DEM rasters to build 30m provincial hazard matrices.

HONORS

Nov 2026

KBTG Kampus Hackathon 2026 – Top 30 Overall & Top 10 Data Science

Pitcher, Developer & Concept Lead · Conceived the K-NEXUS product architecture and delivered the stage pitch at Kasikorn Business-Technology Group.

Dec 2025

AIRA & AIFUL / Aihack Thailand 2025 – National Finalist (Team OEIA)

ML & Financial Risk Lead · Pitched credit-cutoff vs. interest-margin profitability trade-offs to executive judges (Chulalongkorn Business School × ProbSpace).

ASSESSMENTS

Mar 2026

SKILLKAMP by KBTG – Data Analyst Assessment (STAMP)

Verified score of 97/150 (Above KBTG Industry Benchmark of 90) · Data Cleaning 23/30, EDA & Visualization 20/30, Foundations 20/30, ML 18/30.

Mar 2026

SKILLKAMP by KBTG – Digital & Performance Marketer (STAMP)

Verified score of 98/150 (Above KBTG Industry Benchmark of 94) · Digital Marketing Analytics 26/30 (Benchmark: 17.94), Data-Driven Marketing 21/30.

Jun 2026

Oxford Placement Test – English Proficiency (AUA / Oxford University Press)

Verified CEFR B1 (Overall Score: 51/120 · Use of English: 54 / B1, Listening: 48 / B1).